

Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment Name:

Experiment # 1: Familiarization with Fundamental Logic Gates

Experiment # 2: Universal Gates, Applications of Boolean Algebra

Experiment # 3: Parity Bit Checker and Generator

Required Components for Experiment -1,2,3:

1. IC 7408 × 1
2. IC 7432 × 1
3. IC 7404 × 1
4. IC 7400 × 1
5. IC 7402 × 1
6. IC 7486 × 1
7. IC 4077 × 1

Experimental Setup:

*Attach the signed circuit diagram or experimental setup part from the lab

Experimental Setup for Experiment - 1,2,3:

Result:

*Attach the signed result part from the lab

Result for Experiment - 1,2,3:

(You need to answer only the following two questions for report 1)

Answer the following questions also as part of discussion:

1. Implement the following function using NAND gate only:
 $(A+(AB+C'D))'$. Do not simplify the function. Draw the circuit diagram only.
2. Draw the circuit diagrams of 3-bit parity checker and generator using NOR gate(s) only